

incompressible uncertainty that, in theory, will defy science and forever remain an uncertainty. This minimum uncertainty was discovered by Werner Heisenberg in 1927. I find it ludicrous to present the uncertainty principle as having anything to do with uncertainty. Why? First, this uncertainty is Gaussian. On average, it will disappear—recall that no one person’s weight will significantly change the total weight of a thousand people. We may always remain uncertain about the future positions of small particles, but these uncertainties are very small and very numerous, and they average out—for Pluto’s sake, they average out! They obey the law of large numbers we discussed in [Chapter 15](#). Most other types of randomness do not average out! If there is one thing on this planet that is not so uncertain, it is the behavior of a collection of subatomic particles! Why? Because, as I have said earlier, when you look at an object, composed of a collection of particles, the fluctuations of the particles tend to balance out.

But political, social, and weather events do not have this handy property, and we patently cannot predict them, so when you hear “experts” presenting the problems of uncertainty in terms of subatomic particles, odds are that the expert is a phony. As a matter of fact, this may be the best way to spot a phony.

I often hear people say, “Of course there are limits to our knowledge,” then invoke the greater uncertainty principle as they try to explain that “we cannot model everything”—I have heard such types as the economist Myron Scholes say this at conferences. But I am sitting here in New York, in August 2006, trying to go to my ancestral village of Amioun, Lebanon. Beirut’s airport is closed owing to the conflict between Israel and the Shiite militia Hezbollah. There is no published airline schedule that will inform me when the war will end, if it ends. I can’t figure out if my house will be standing, if Amioun will still be on the map—recall that the family house was destroyed once before. I can’t figure out whether the war is going to degenerate into something even more severe. Looking into the outcome of the war, with all my relatives, friends, and property exposed to it, I face *true* limits of knowledge. Can someone explain to me why I should care about subatomic particles that, anyway, converge to a Gaussian? People can’t predict how long they will be happy with recently acquired objects, how long their marriages will last, how their new jobs will turn out, yet it’s subatomic particles that they cite as “limits of prediction.” They’re ignoring a mammoth standing in front of them in favor of matter even a microscope would not allow them to see.

Can Philosophers Be Dangerous to Society?

I will go further: people who worry about pennies instead of dollars can be